

Causal Normalizing Flows: From Theory to Practice

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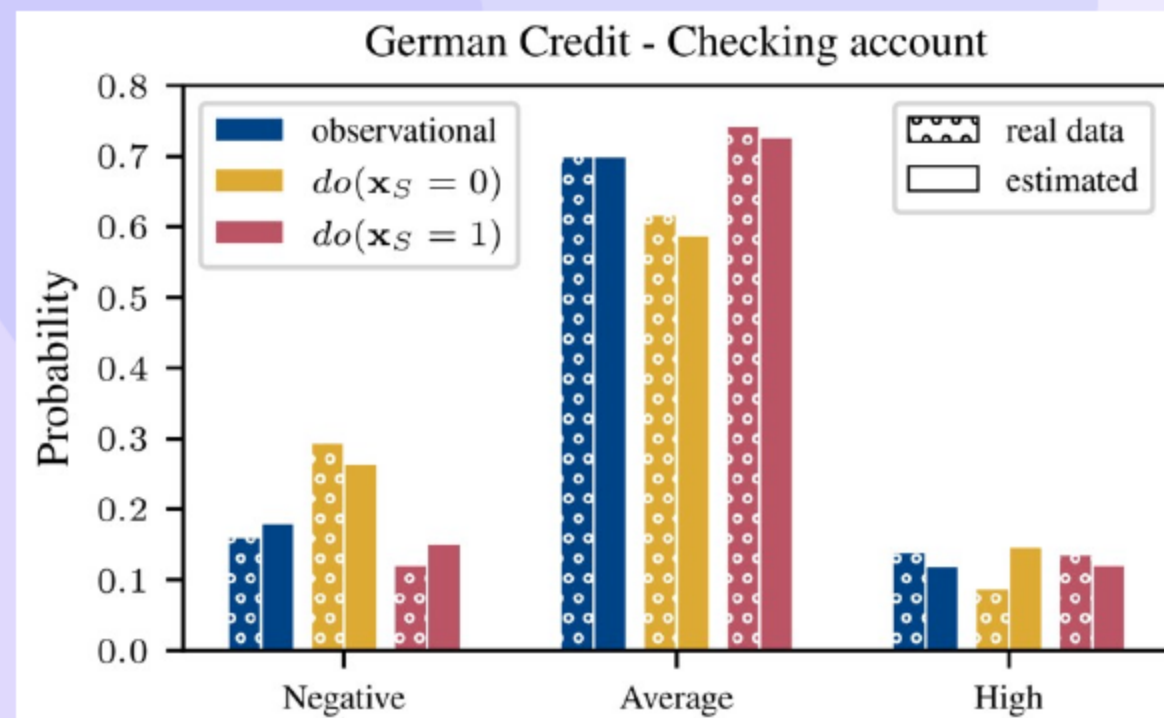
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Causal Normalizing Flows: From Theory to Practice — Authors and Affiliations

- Authors
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 - Isabel Valera
- Affiliations
 - 1 Saarland University in Computer Science, Germany
 - 2 Max Planck Institute for Intelligent Systems in Tübingen
 - 3 Max Planck Institute for Software Systems in Saarbrücken
- Presentation focus on [single-flow](#) causal modeling with normalizing flows

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Real-world Motivation and Failure Modes



- Goal is to estimate interventional effects from observational data with partial causal structure
- Multi-network approaches suffer from error accumulation along long causal paths
- Large and hard to train parameter counts hinder robust learning in practice
- Implicit densities prevent exact likelihoods and principled interventions
- German Credit use case motivates fairness auditing with partial graphs
- Real impact: audits for **fair** credit decisions and [reliable](#) what-if analysis

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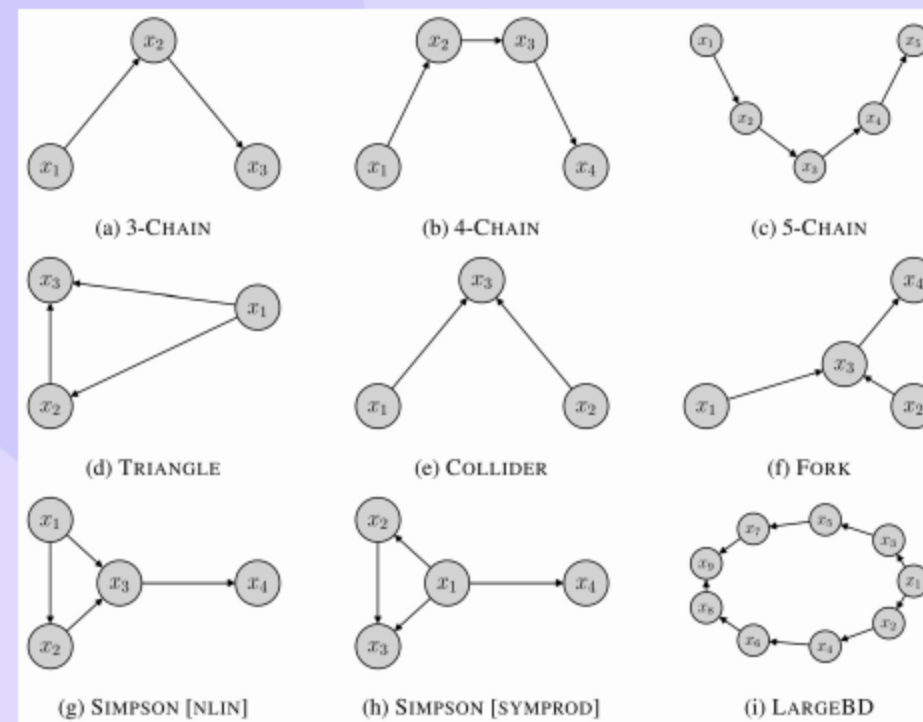
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Why Single-Flow Causal Modeling (our preview)

- Single causal normalizing flow learns the entire data generating process
 - Theoretical support for broad structural causal models
 - Causal dependencies enforced via architecture and partial graphs
 - Efficient and sound intervention mechanism within the flow
 - Support for mixed data types with [improved performance](#)
- Design enables exact likelihoods and consistent interventions
- Empirical results outperform competing single network baselines

Positioning vs Prior Work (with citations)

- Comparison includes CAREFL [18], VACA [36], GNN-SCM [43], and diffusion-based counterfactuals [3]
- Our approach strengthens links between causal models and autoregressive flows with principled masking
- Our method achieves broader identifiability, higher efficiency, and exact likelihoods
 - Sound intervention mechanism provides [correct interventions](#)
- See refs: [1,3,12,18,28,36,43]



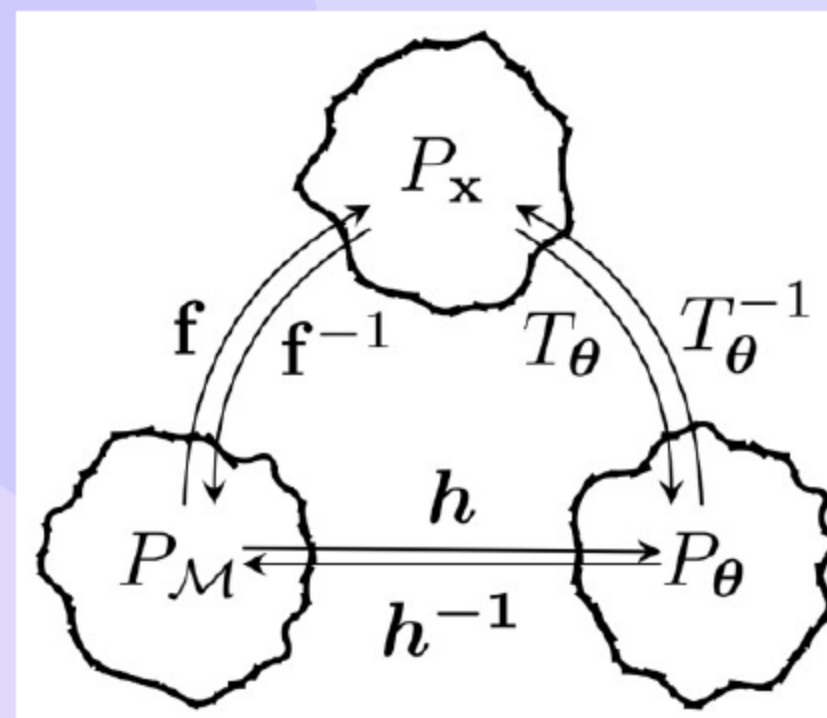
- Structural causal models map independent exogenous variables to endogenous variables through component functions
- Causal graphs define parents and ancestors under an acyclic ordering
- Queries include interventional distributions and counterfactual outcomes
- Structural equivalence compares zero patterns and relative sparsity

NFs and Autoregressive Structure

- Normalizing flows model densities through invertible transforms with tractable likelihoods
- Autoregressive normalizing flows use ordering respecting conditioners with triangular Jacobians
- Strictly monotone transformers enable stable optimization and exact determinants
 - Triangular structure leads to **efficient computation**

Problem and Assumptions

- Objective is to learn an autoregressive normalizing flow from observational data with partial causal knowledge
- Assumption requires an invertible and differentiable data generating process
- Assumption imposes an acyclic causal graph without feedback
- Assumption enforces causal sufficiency with independent exogenous variables



- Structural models and autoregressive flows align within triangular monotone increasing maps with factorized noise
- Autoregressive layers approximate this map class through universal monotone transforms
- Identifiability ensures generators differ only by invertible component wise transforms
 - Uniqueness up to elements supports **stable interventions**

Causal Consistency and Implications

- Matching the observational distribution implies coincident functional dependencies
- Causal consistency links Jacobian sparsity to the adjacency structure in both directions
- Shared map class yields uniqueness up to element wise transforms and preserved paths
 - Consistency supports **reliable counterfactuals**

Design Pitfalls and Graph Masking

- Generative direction requires masking with identity plus adjacency to prevent shortcuts
- Depth must exceed the graph diameter to separate direct and indirect paths
 - Insufficient depth risks **spurious dependencies**
- Abductive direction uses masking with identity minus adjacency to capture indirect effects
 - Single layer captures all relevant dependencies by design

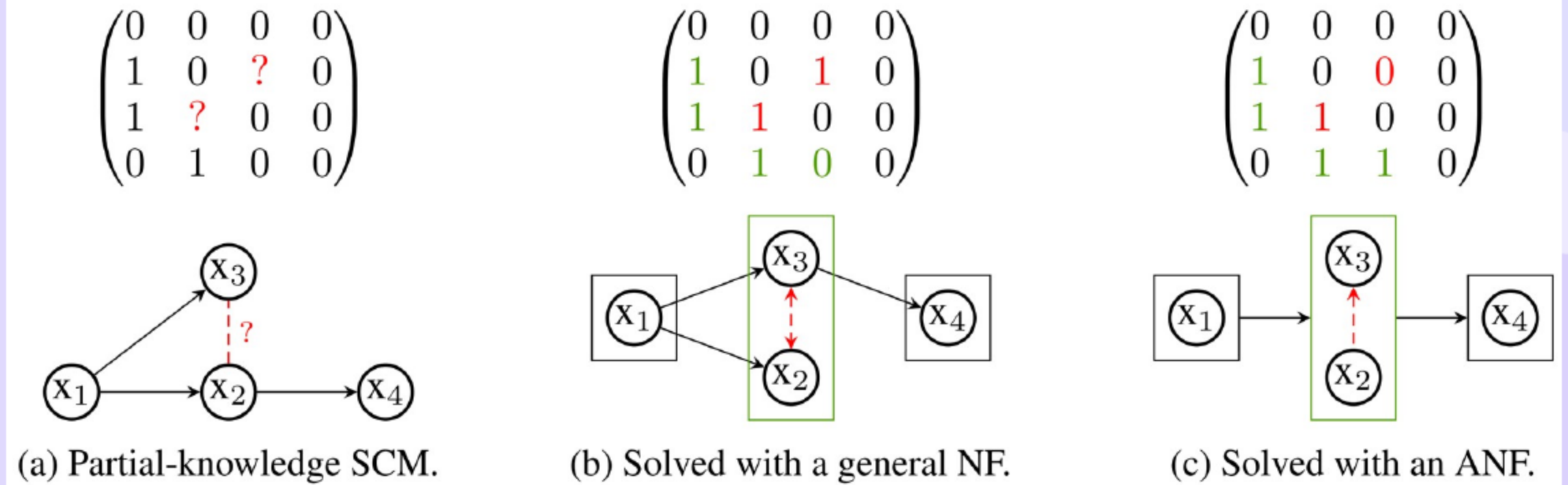
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Preferred Design: Single-Layer Abductive

- Single masked layer in the abductive direction is expressive and causally consistent
 - Universal triangular monotone approximator enables **compact models**
- Causal consistency matches sparsity patterns at the optimum for both directions
- Regularization with Jacobian sparsity helps when the graph is known but unenforced
 - Penalty discourages **spurious influences**

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Practical Scope: Discrete + Partial Graphs



- Mixed data are supported by noise augmentation for discrete variables without losing identifiability
- Partial graphs are handled by contracting strongly connected components into an acyclic block graph
- Block level reasoning enables training with incomplete structural knowledge
 - Generalization yields **robust applicability**

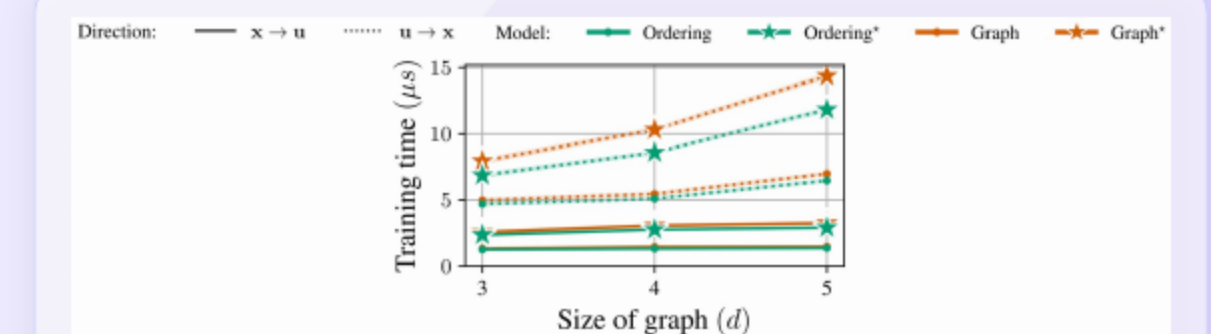
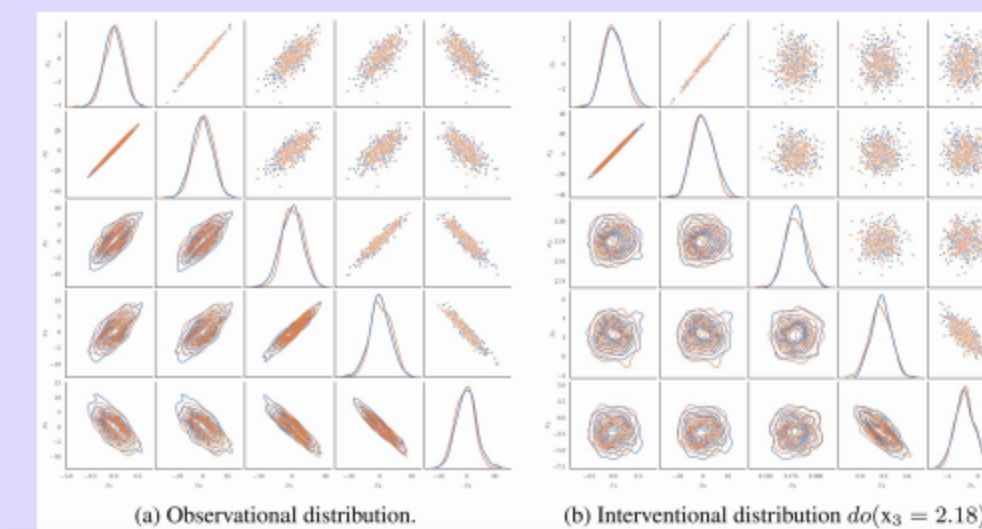
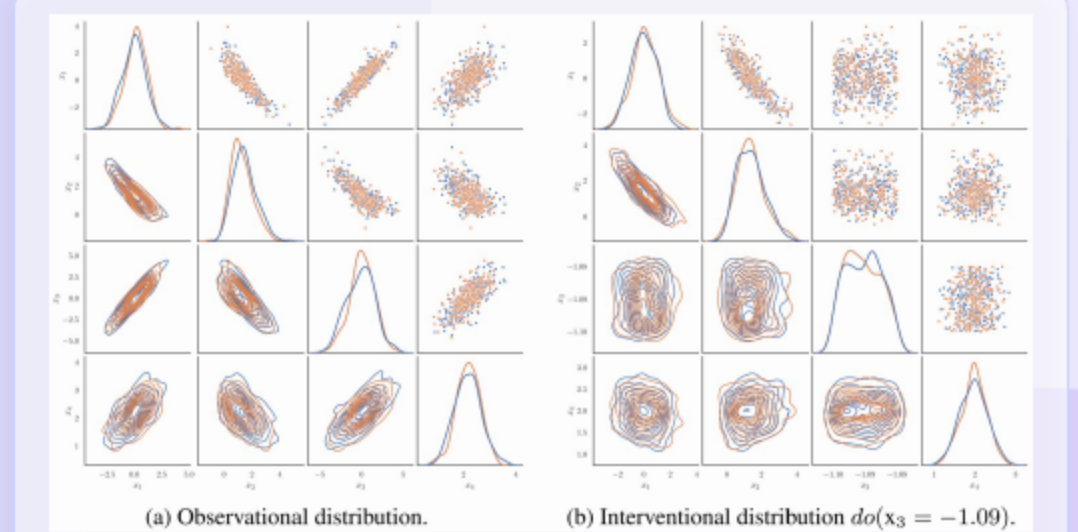
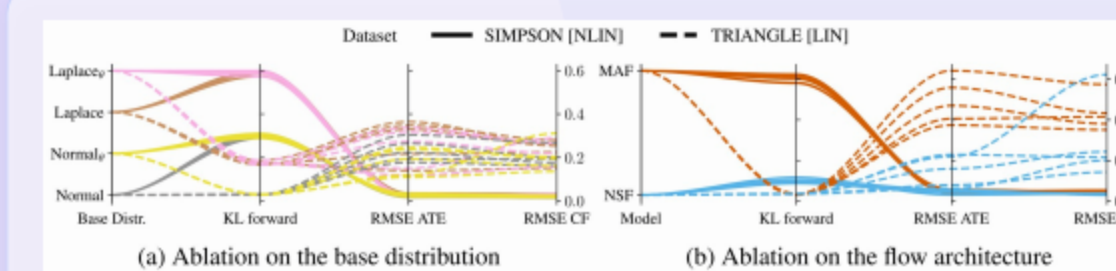
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Do-Operator: Concept + Procedure

- Interventions are implemented by modifying the exogenous distribution while preserving structural equations
- Procedure uses the learned transform to compute exogenous variables and set intervened components
- Approach supports counterfactual backtracking and non recursive structures when plausible
 - Built in mechanism ensures **sound interventions**

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Key Empirical Findings (Ablations summary)



- Baselines: **CAREFL** [18], **VACA** [36], **GNN-SCM** [43], **diffusion-based counterfactuals** [3]
- See refs: [1,3,12,18,28,36,43]

12 Comparative Evaluation vs CAREFL (AISTATS'21) [18]

- ATE RMSE ↓ x% (avg. across datasets)
- CF RMSE ↓ y% (avg. across datasets)
- Evaluation speed ↑ z× (avg. across datasets)
- CAREFL (Sánchez-Martín, Javaloy, Valera, AISTATS 2021) [18]

[18]; [36]

13 Use-Case: Counterfactual Fairness on German Credit

- Setup following Sánchez-Martín et al. [36]; partial graph per Chiappa [4]. Abductive single non-affine spline layer.
- Classifier on exogenous variables without the sensitive component achieves counterfactual fairness
 - Fair variant maintains **competitive accuracy**
- Unfairness ranking follows expectations with full models being most unfair
 - Removal of sensitive paths reduces **disparate impact**

[9]; [36]; [4]

14 Conclusions, Limitations, Future Work

- Causal normalizing flows provide principled modeling with a sound intervention mechanism and support for mixed data
 - Unified design yields **consistent reasoning**

Thank You

Questions & Discussion